Rutuj Kale

Data Science and AI/ML Intern | Python | Machine Learning | AI Evaluation

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Professional Summary

Artificial Intelligence and Data Science undergraduate with hands-on project experience in Python development, machine learning, data analytics, and model evaluation. Experienced in processing structured datasets, investigating model performance, and developing applications involving document extraction and computer vision. Familiar with Pandas, NumPy, scikit-learn, PyTorch, SQL, PostgreSQL, and Git/GitHub. Interested in applying AI and intelligent automation to practical business problems, with a focus on model reliability, analytical problem-solving, and measurable outcomes.

Education

B.Tech. in Artificial Intelligence and Data Science **Expected 2027**

Jawaharlal Nehru Engineering College, MGM University **CGPA: 7.77/10**

Relevant Coursework: Machine Learning, Data Structures and Algorithms, Database Management Systems, Java Programming, Cloud Computing.

Technical Skills

Programming Languages: Python, SQL, Java
Data Science: Pandas, NumPy, data cleaning, preprocessing, exploratory data analysis, data transformation
Machine Learning: scikit-learn, PyTorch, classification concepts, model evaluation, performance metrics
AI and Development: NLP fundamentals, computer vision, YOLO, FastAPI, REST API concepts
Databases: PostgreSQL, MySQL, relational data, SQL queries and structured data management
Tools and Libraries: Git, GitHub, Jupyter Notebook, Docker fundamentals, PyMuPDF, Tkinter, Edge-TTS
Core Practices: Debugging, analytical problem-solving, modular development, data validation fundamentals

Technical Projects

FraudLens – Machine Learning Evaluation Experiment *Python | Machine Learning | Evaluation*

- Investigated how a deliberately flawed evaluation setup can make a fraud-detection model appear more effective than it actually is, demonstrating the importance of reliable model evaluation.
- Observed PR-AUC change from 0.9560 to 0.9927 under the broken evaluation setup, highlighting the risks of misleading metrics and the need for careful validation and interpretation.
- Used the experiment to examine evaluation methodology and communicate why model quality cannot be judged by a single performance score without context.

Blockchain Business Intelligence and DeFi Analytics *Python | Pandas | SQL | PostgreSQL*

- Collected, cleaned, and transformed public Aave V3 Polygon transaction data into structured records stored in PostgreSQL for downstream analysis.
- Explored wallet segmentation, cohort retention, and engagement patterns to investigate user activity and behavioural trends across transaction data.
- Examined limitations of active-wallet counts as a standalone engagement metric, using analytical reasoning to interpret results in their business context.

Smart Traffic Management System *Python | YOLO | Computer Vision*

- Worked on traffic-video data collection, cleaning, and preprocessing for a YOLO-based vehicle-detection workflow.
- Explored computer-vision-based traffic analysis and data-processing approaches for vehicle-detection workloads.
- Contributed to a project focused on applying AI techniques to practical traffic-management challenges.

PDF Audiobook Generator *Python | PyMuPDF | Edge-TTS | Tkinter*

- Developed a Python desktop application that extracts text from PDF documents, converts the content into speech, and provides playback controls through a graphical interface.
- Integrated document processing, speech generation, and background execution to help maintain application responsiveness during longer operations.

UpCrew AI – Multimodal Talent Intelligence Platform *Python | FastAPI | NLP*

- Developing a platform concept that combines text, audio, and visual inputs for structured candidate assessment.
- Exploring modular integration of NLP and computer-vision components through a Python API architecture.

Achievements and Technical Activities

- **Smart India Hackathon 2024:** College-round participant; Fixed-Wing Drone for Emergency Response.
- **Smart India Hackathon 2025:** Participant; Gamified Disaster Preparedness and Risk Reduction Platform.
- **Google Solution Challenge 2025:** Participant; Smart Traffic Management System.